



Characterization of the telomerase modulating activities of yeast DNA helicases

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Abstract

Eukaryotes with linear chromosomes circumvent the end replication problem via the action of a specialized ribonucleoprotein reverse transcriptase known as telomerase. Cells lacking telomerase activity will senesce when their chromosome ends shorten to a critical length. In contrast, cancer cells can become immortalized by upregulating telomerase to lengthen telomeres during each cycle of DNA replication. Thus, the regulation of telomerase is critical for normal telomere homeostasis. Of the various known ways that telomerase activity is modulated *in vivo*, recent studies have demonstrated that DNA helicases are involved. In *Saccharomyces cerevisiae*, the Hrq1 and Pif1 helicases act in a pathway that regulates telomerase extension at telomeres and at DNA double-strand DNA breaks. *In vitro* analysis demonstrates that when these helicases are combined in reactions, they synergistically inhibit or stimulate telomerase activity depending on which helicase is catalytically active. Here, we describe the methods for the overproduction and purification of Hrq1 and Pif1. We also report the preparation of partially purified cell extracts with telomerase activity and how the effects of these helicase on telomerase activity can be assessed *in vitro*.



1. Introduction

Due to the semi-discontinuous nature of DNA replication, the duplication of linear chromosomes must overcome the end replication problem, i.e., the progressive shortening of chromosome ends as the RNA primers for terminal Okazaki fragment synthesis are removed (Bonnell, Pasquier, & Wellinger, 2021). Prokaryotes with linear chromosomes have developed several ways to deal with this challenge (Casjens, 1999; Kobryn & Chaconas, 2001; Volff & Altenbuchner, 2000; Yang et al., 2002). However, a single conserved mechanism is utilized by most eukaryotic genomes. Organisms from yeast to humans have evolved specialized chromosome ends called telomeres and a dedicated ribonucleoprotein complex called telomerase that together protect chromosome ends from degradation and solve the end replication problem (Bonnell et al., 2021).

In the majority of studied eukaryotes, telomeric DNA is comprised of short repeats of TG-rich sequences, such as TG₁₋₃ in *Saccharomyces cerevisiae* and TTAGGG in *Homo sapiens* (Bonnell et al., 2021). The composition of such sequences can lead to the formation of inter- and intramolecular G-quadruplex structures in the context of single-stranded DNA (Bochman, Paeschke, & Zakian, 2012), which is likely a protective mechanism to inhibit nuclease activity from degrading naked telomeric DNA (Smith et al., 2011). Under most circumstances though, telomeric DNA is highly proteinaceous, being chromatinized by a number of transcriptional silencing proteins and sequence-specific DNA binders (Bonnell et al., 2021). This protects telomeres from degradation and being recognized as DNA double-strand breaks.

The telomerase holoenzyme is comprised of several subunits, including an RNA that acts as the template for a reverse transcriptase, with other proteins providing regulatory or structural roles. These factors in *S. cerevisiae* are the *TLC1* RNA, Est2 reverse transcriptase, and the Est1 and Est3 proteins, respectively (Wellinger & Zakian, 2012). However, many other proteins bind to the large *TLC1* molecule, and the holoenzyme interacts with many other proteins, forming a mega-complex in vivo (Lin et al., 2015). This array of protein–protein and protein–nucleic acid interactions, as well as a complex set of post-translational modifications of the protein players involved, regulate telomerase to avoid problems associated with too much or too little telomerase activity (Bonnell et al., 2021; Wellinger & Zakian, 2012).

It is known that many DNA helicases localize to telomeres and perform roles in DNA replication, recombination, and repair that are necessary for telomere maintenance (Bochman, 2014). One well-studied example is the *S. cerevisiae* Pif1 helicase, which is a catalytic inhibitor of telomerase (reviewed in Bochman, Sabouri, & Zakian, 2010). Evidence suggests that Pif1 uses its helicase activity to unwind the telomeric DNA-*TLC1* RNA hybrid formed by telomerase and thereby evict telomerase from its sites of action (Boule, Vega, & Zakian, 2005). In contrast, the second PIF1 family helicase in budding yeast, Rrm3, is not known to directly affect telomerase but instead is necessary for efficient DNA replication through telomeric DNA (Ivessa, Zhou, Schulz, Monson, & Zakian, 2002). Similarly, RecQ family helicases, such as Sgs1 in *S. cerevisiae*, can participate in telomerase-independent telomere maintenance via DNA recombination (Azam et al., 2006; Cohen & Sinclair, 2001; Huang et al., 2001; Johnson et al., 2001).

Recently, it has been shown that the second RecQ helicase in *S. cerevisiae*, Hrq1 (a homolog of human RECQL4), is also involved in multiple facets of telomere maintenance (Bochman, Paeschke, Chan, & Zakian, 2014). For instance, telomere addition to double-strand breaks increases in the absence of Hrq1, Hrq1 is necessary for Type I survivor formation in cells lacking telomerase, and telomeres hyper-lengthen in *hrq1Δ pif1* double mutants. These effects also appear to be direct, as Hrq1 binding is enriched at telomeres in vivo. Using biochemistry, we have further characterized the synergistic in vitro activity of recombinant Hrq1 and Pif1 to regulate telomerase extension of a model telomeric DNA substrate (Nickens, Rogers, & Bochman, 2018). Hrq1 alone stimulates telomerase activity at high helicase concentrations, Pif1 alone stimulates telomerase at low helicase concentrations but inhibits its activity at high concentrations, and the combination of helicases significantly inhibits telomerase at all concentrations tested (Fig. 1A) (Nickens et al., 2018). By using either catalytically inactive Hrq1-K318A or Pif1-K264A recombinant helicase in the dual-helicase reactions, the combination either acts as a strong inhibitor of telomerase (Fig. 1B) or a strong stimulator of telomerase (Fig. 1C). We hypothesize that the cell regulates Hrq1 and Pif1 activity—and thus their regulation of telomerase—to establish a homeostatic telomere length.

The methods used to perform the in vitro telomerase activity assays with and without added helicase(s) are described in detail below. However, the basic assay is amenable to many modifications. One could assess the effects of different DNA sequences and structures on telomerase,

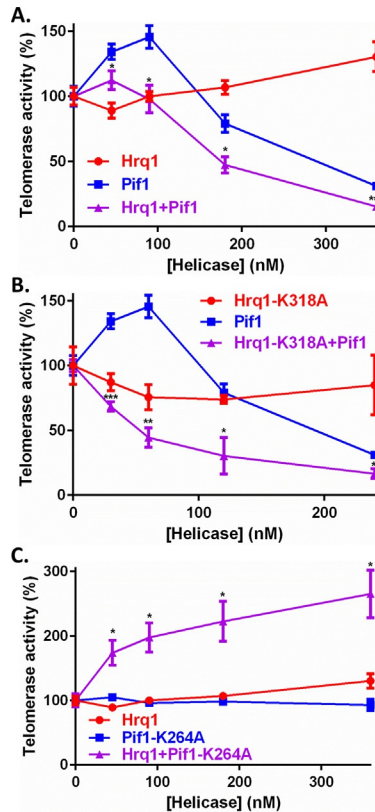


Fig. 1 The effects of Hrql and/or Pif1 on in vitro telomerase activity. (A) The wild-type helicases alone display differing effects on telomerase, but in combination, they inhibit telomerase primer extension significantly better than Pif1 alone. (B) The inactive Hrql-K318A mutant has no effect on telomerase activity alone but, together with wild-type Pif1, is a potent telomerase inhibitor. (C) The inactive Pif1-K264A mutant has no effect on telomerase activity alone but, in the presence of wild-type Hrql, greatly stimulates telomerase activity. * $P < 0.01$, ** $P < 0.001$, and *** $P < 0.0001$. This figure was adapted from Nickens, D. G., Rogers, C. M., & Bochman, M. L. (2018). The *Saccharomyces cerevisiae* Hrql and Pif1 DNA helicases synergistically modulate telomerase activity in vitro. *The Journal of Biological Chemistry*, 293(37), 14481–14496. doi:10.1074/jbc.RA118.004092.

screen small molecules for impacts on the reaction, test different or additional helicases, and/or determine the effects of other recombinant proteins known or suspected to be involved in telomere maintenance.



2. Before you begin

Like many biochemical methods, measuring telomerase activity in vitro is frontloaded. It can take quite a while to generate and assemble

all of the necessary reagents, but once that is accomplished, data can quickly be generated. In our experience, it is best not to rush the preparation of any reagent and to, for instance, perform DNA unwinding assays with the recombinant helicases to verify their activity before moving forward with telomerase assays. We have also found that careful buffer preparation and titration of each telomerase preparation is essential for repeatable experiments. Thus, we recommend reserving 10% of any reagent (protein, buffer, etc.) before completely depleting the stocks in the event that subsequent preparations of reagents produce variable results. Having samples of the original materials on hand will enable troubleshooting to determine the root of the problem. The following are additional tips and tricks that should be considered before performing the methods described below:

1. After cells cultures are harvested, chill flasks and centrifuge bottles on ice prior to the centrifugation steps. Similarly, when thawing harvested cells for purification, have all equipment cleaned and ready to use, keeping all tubes and centrifuge bottles at 4 °C.
2. Before grinding yeast “BBs,” add and grind dry ice to the grinder mill to chill it before processing cells.
3. Telomerase preparations can take up to 11 h to complete, but it is best to finish the process as quickly as possible in 1 day to preserve activity. Two people sharing part of the processing time can be helpful.
4. Although the previously published versions of the *in vitro* telomerase primer extension assay (Boule et al., 2005; Cohn & Blackburn, 1995) work to assess the effects of the Pif1 helicase on telomerase activity, we found that they were insufficient for examining the effects of Hrq1 alone or in combination with Pif1 (Nickens et al., 2018; Nickens, Sausen, & Bochman, 2019). This is because the Hrq1 helicase has a larger single-stranded DNA binding site size than Pif1, necessitating a longer oligonucleotide substrate in the reactions. Thus, substrate design and selection should reflect the alterations that one may wish to make with this assay, such as including a different helicase.



3. Materials and equipment

3.1 General equipment

- Vertical polyacrylamide gel electrophoresis equipment (CBS Scientific Vertical Mini-gel System)
- Large vertical gel electrophoresis equipment (Bethesda Research Laboratories, Model S2)
- Heated water baths: 30 and 42 °C

- Standing incubators: 27, 30, and 37 °C
- Shaking incubators capable of heating ($\geq 37^\circ\text{C}$) and cooling ($\leq 18^\circ\text{C}$)
- Centrifuges (micro, table-top, super-speed, and ultra) capable of cooling to 4 °C
- Sonicator or alternative method for bacterial cell lysis
- Medium-pressure liquid chromatography instrument (AKTA Pure)
- Liquid nitrogen Dewar
- Cell grinder (electric coffee/spice grinder mill)
- Phosphorimager plate
- Phosphorimager
- pH meter

3.1.1 Strains

- *S. cerevisiae* strain JBBY26, a derivative of BCY123 (*MATa can1 ade2 trp1 ura3-52 his3 leu2-3,112 pep4::HIS3 prb1::LEU2 bar1::HISG lys2::pGAL1/10-GAL4*) (Zhao et al., 1994) harboring a dual over-expression plasmid for *TLC1* and *Est2* (Boule et al., 2005)
- *Escherichia coli* strain Rosetta 2(DE3)pLysS ($F^- ompT hsdS_B(r_B^- m_B^-) gal dcm$ (DE3) pLysSRARE2 (Cam^R))
- *Spodoptera frugiperda* cell line Sf9

3.1.2 Reagents and chemicals

- Protease inhibitor mix: 600nM leupeptin, 2 μM pepstatin A, 2mM benzamidine, and 1mM phenylmethanesulfonyl fluoride (PMSF). Commercially available protease inhibitor mixes are also appropriate
- Urea
- NaCl
- Glycerol
- Lysozyme
- β -mercaptoethanol
- HEPES
- EDTA
- Tris
- HCl
- NaOH
- NaOAc
- MgCl_2
- Triton-X100
- NP-40 or NP-40 substitute

- $\text{Mg}(\text{OAc})_2$
- $\text{Zn}(\text{OAc})_2$
- NH_4OAc
- Dithiothreitol (DTT)
- Tween-20
- Spermidine
- dGTP
- dTTP
- RNase inhibitor, murine (New England BioLabs)
- SDS
- Proteinase K
- $[\alpha\text{-}^{32}\text{P}]\text{TTP}$ ($10\text{ }\mu\text{Ci}/\mu\text{L}$; Perkin-Elmer)
- $[\gamma\text{-}^{32}\text{P}]\text{ATP}$ (Perkin-Elmer)

3.1.3 Homemade buffers

- Pif1 Lysis Buffer: 50mM sodium phosphate buffer (pH7.5), 300mM NaCl, 5mM β -mercaptoethanol, 10% (w/v) glycerol, protease inhibitor mixture, and 0.5mg/mL lysozyme
- Pif1 IEX Buffer: 50mM sodium phosphate buffer (pH6.8), 10% (w/v) glycerol, 0.5mM EDTA (pH8.0), and 4mM β -mercaptoethanol
- Pif1 Storage Buffer: 25mM HEPES (pH7.5), 150mM NaCl, 2mM β -mercaptoethanol, 0.1mM EDTA (pH8.0), and 30% (w/v) glycerol
- L Buffer: 40mM Tris-HCl (pH8.0), 500mM NaOAc, 1.1mM MgCl_2 , 0.1mM EDTA, 0.1% (w/v) Triton X-100, 0.2% (w/v) NP-40, 10% (w/v) glycerol, and 0.1mM PMSF
- TMG30 Buffer: 10mM Tris-HCl (pH8.0), 30mM NaOAc, 2.2mM MgCl_2 , 0.1mM EDTA, 0.1% (w/v) Triton X-100, 0.2% (w/v) NP-40, 10% (w/v) glycerol, and 0.1mM PMSF
- TMG500 Buffer: 10mM Tris-HCl (pH8.0), 500mM NaOAc, 2.2mM MgCl_2 , 0.1mM EDTA, 0.1% (w/v) Triton X-100, 0.2% (w/v) NP-40, 10% (w/v) glycerol, and 0.1mM PMSF
- TMG900 Buffer: 10mM Tris-HCl (pH8.0), 900mM NaOAc, 2.2mM MgCl_2 , 0.1mM EDTA, 0.1% (w/v) Triton X-100, 0.2% (w/v) NP-40, 10% (w/v) glycerol, and 0.1mM PMSF
- Hrq1 Lysis Buffer: 50mM Na-HEPES (pH7.6), 0.4M NaCl, 20mM imidazole, and 10% (w/v) glycerol
- Hrq1 Storage Buffer: 25mM Na-HEPES (pH8.0), 30% (w/v) glycerol, 300mM NaOAc (pH7.6), 25mM NaCl, 5mM $\text{Mg}(\text{OAc})_2$, 1mM DTT and 0.1% (v/v) Tween-20

- Telomerase Reaction Buffer: 20 mM Tris–HCl (pH 8.0), 20 mM NaCl, 5 mM MgCl₂, 1 mM DTT, 1 mM spermidine, 1 μ L of 10 μ M telomeric primer, 50 μ M dGTP, 5 mM TTP, 1 μ L of [α -³²P]TTP (10 μ Ci/ μ L), RNase inhibitor (1 U/ μ L) and 1–3 μ L of telomerase-enriched extract
- Stop Buffer: 20 mM Tris–HCl (pH 8.0), 1 mM EDTA, 0.5% (w/v) SDS, 250 μ g/mL proteinase K, and $\sim$ 1000 cpm of [γ -³²P]ATP-labeled loading control oligonucleotide
- Formamide load buffer: 10 mM NaOH, 95% formamide, 0.05% bromophenol blue, and 0.05% xylene cyanol

3.1.4 Oligonucleotides

- Tel15: 5′-TGTGGTGTGTGTGGG-3′. This is the standard substrate used to standardize telomerase activity levels and to determine the effects of Pif1 on telomerase activity (Boule et al., 2005).
- Tel30: 5′-CGCCATGCTGATCCGTGTGGTGTGTGTGGG-3′. This substrate can be used with helicases such as Hrq1 that have binding footprints >15 nt and/or in telomerase displacement assays (Nickens et al., 2018).
- Tel50: 5′-GTGTGGGTGTGGTGTGGGTGTGGTGTGGGTGTGTGGTGTGGTGTGTGTGG-3′. This substrate is used to determine the effects of Hrq1 or the combination of Hrq1 and Pif1 on telomerase.
- Loading control: We used ³²P-labeled Tel15 as the loading control for telomerase assays with Tel30 or Tel50 as the extended primers. For assays in which Tel15 was the extended primer, we used ³²P-labeled Tel50 as the loading control.



4. Step-by-step method details

4.1 Over-expression and purification of the nuclear isoform of Pif1

Despite being a large eukaryotic protein with an N-terminal domain that is predicted to be natively disordered (Andis, Sausen, Alladin, & Bochman, 2018), *S. cerevisiae* Pif1 is amenable to over-expression and purification from *E. coli*. Many groups have successfully generated recombinant Pif1 in this manner, and we favor the expression constructs and basic protocols of the Raney lab, which produce excellent yields of the nuclear isoform of Pif1 with no affinity tags (Ramanagoudr-Bhojappa, Blair, Tackett, & Raney, 2013). The catalytically inactive Pif1-K264A mutant can be purified in

the same manner, as well as various other substitution and domain truncation mutants (Nickens et al., 2018, 2019; Ononye, Sausen, Balakrishnan, & Bochman, 2020).

1. Transform the expression strain of *E. coli* with plasmid pSUMO-Pif1 (Ramanagoudr-Bhojappa et al., 2013). We have achieved excellent over-expression with both Rosetta 2(DE3) pLysS cells (Novagen) and NiCo21(DE3) cells (New England Biolabs) that also carry the pLysS plasmid. Select for transformants on LB medium supplemented with the appropriate antibiotics, e.g., 50 µg/mL kanamycin and 34 µg/mL chloramphenicol for the above-suggested strains, at 37 °C.
2. Inoculate 5-mL of LB medium supplemented with the same antibiotics with a fresh transformant colony, and grow overnight at 30 °C with aeration. Only fresh transformants should be used as we have observed little-to-no expression from transformants stored on plates for extended periods of time or from glycerol stocks of expression strains carrying the pSUMO-Pif1 plasmid.
3. Back-dilute the overnight culture into 2x YT medium supplemented with 10mM MgSO₄, 1mM Zn(OAc)₂, and the same antibiotics listed above, and grow the cells to mid-log phase at 24 °C with aeration. A 1-L culture is generally sufficient to produce 0.5 mg of purified protein.
4. Induce SUMO-Pif1 over-expression by adding IPTG to a final concentration of 0.2mM, and incubate the cells for 14–18 h at 18 °C with aeration. All subsequent steps should be performed at 4 °C or on ice using instruments and equipment that is likewise chilled.
5. Harvest the cells by pelleting in a centrifuge at 7000 × *g* for 10 min, decant the growth medium, and resuspend the cells in Lysis Buffer.
6. Lyse cells, and subject the lysate to ultracentrifugation at 150,000 × *g* for 1 h. We have used a Fisher Scientific FB505 sonicator, a Microfluidics microfluidizer, an Avestin EmulsiFlex-C3 emulsifier, and a high-pressure cell cracker to lyse cells with comparable results, so any lysis method that does not introduce excessive amounts of heat is likely to be acceptable.
7. Apply the ultracentrifugation supernatant to a 5-mL Talon HiTrap column, wash with 10 column volumes (CVs) of Lysis Buffer, 20 CVs of Lysis Buffer supplemented with 15mM imidazole, and elute with Lysis Buffer containing 200mM imidazole using an AKTA FPLC or similar medium-pressure liquid chromatography instrument.
8. Identify the peak SUMO-Pif1-containing fractions by SDS-PAGE and Coomassie staining analysis of 10-µL aliquots of each elution fraction,

pool the peak fractions, and dialyze them against 20-fold excess lysis buffer to remove imidazole.

9. Treat the SUMO-Pif1 with Ulp1 SUMO protease to remove the N-terminal SUMO tag, which itself is $6 \times$ His tagged, and reapply to the Talon HiTrap column. The His-tagged Ulp1 protease, uncleaved SUMO-Pif1, and contaminants from the elution in Step 7 will be retained on the Talon column, and untagged Pif1 will elute in the column flow-through.
10. To further purify the untagged nuclear isoform of Pif1 diluting it 1:1 with IEX Buffer and applying it to a Resource S. Elute with a NaCl gradient from 150 mM to 1 M. Once again, identify Pif1-containing fractions by SDS-PAGE and Coomassie staining, and pool the peak fractions.
11. Concentrate the Pif1 with Amicon Ultra-4 centrifugal filters with a 30,000 cutoff, and store at -80°C in Storage Buffer.

4.2 Over-expression and purification of Hrq1

S. cerevisiae Hrq1 is a larger enzyme (~ 125 kDa) than Pif1, and although it can be over-expressed and purified from *E. coli* in a similar manner as that described for Pif1 above (Bochman et al., 2014), the yields are low. Even using the high-density autoinduction method (Studier, 2005), only ~ 100 μg of purified recombinant Hrq1 can be expected from 100 g of cell paste (Fig. 2A). Thus, we typically generate Hrq1 using baculovirus-infected insect cell culture, which has equivalent in vitro activity to Hrq1 over-expressed in *E. coli* but is produced at much higher yields (Rogers et al., 2017; Rogers & Bochman, 2017) (Fig. 2B). The catalytically inactive Hrq1-K318A mutant can be purified in the same manner to that described below for wild-type Hrq1.

1. Infect a 500-mL culture of Sf9 insect cells with Hrq1 baculovirus, and incubate at 27°C for 5 days with gentle aeration.
2. Harvest cells by centrifugation, decant the medium, and snap-freeze the pellet in liquid nitrogen.
3. Thaw the frozen cells (~ 10 g) on ice, and lyse at 4°C by stirring the cell pellet in 100 mL of Hrq1 Lysis Buffer supplemented with fresh 5 mM β -mercaptoethanol, protease inhibitor mix, and 20 $\mu\text{g}/\text{mL}$ DNase I.
4. Clarify the cell lysate was by centrifugation at 4°C for 30 min at 14,000 rpm, and transfer the supernatant into a 50-mL conical tube containing 2.5 mL HIS-Select Nickel Affinity Gel (Sigma) pre-equilibrated with Hrq1 Lysis Buffer.

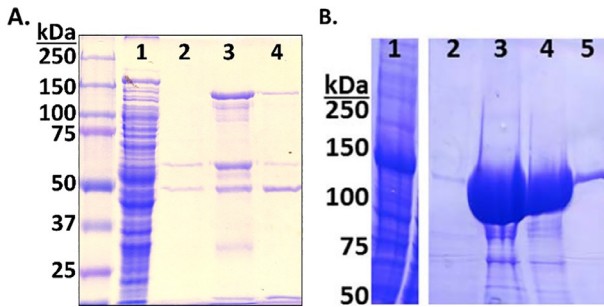


Fig. 2 Generation of recombinant Hrql. Hrql (~125 kDa) was over-expressed in *E. coli* (A) or insect cells (B), and cell lysates were subjected to one step of purification over a Streptactin Sepharose column or in batch mode in the presence of Ni²⁺ affinity resin, respectively. In both images, Lane 1 contains a sample of clarified cell lysate, and the following lanes contain 10- μ L samples of elution fractions. Despite coming from 10-fold less cell mass, the insect cells yielded far more Hrql protein than *E. coli*.

5. Gently mix the contents of the tube for 30 min at 4°C. This can be accomplished with a nutator, rocking platform, tube roller/rotisserie, or an orbital shaker used at low speed.
6. Centrifuge the lysate and resin at 2000 $\times g$ for 5 min at 4°C, and decant most of the supernatant. Then, transfer the remaining supernatant and resin slurry to a 30-mL gravity column, and allow the resin to settle as the supernatant flows through.
7. Wash the resin three times (~30 mL each) with Hrql Lysis Buffer, and elute 10 $\times$ His-Hrql with 5 CVs of Hrql Lysis Buffer containing 500 mM imidazole.
8. Pool the purest fractions, as judged by SDS-PAGE and Coomassie staining, and dialyze against Hrql Storage Buffer.

4.3 Preparation of telomerase extract

To date, no one has purified biochemically useful amounts of recombinant *S. cerevisiae* telomerase holoenzyme (i.e., the Est1-3 proteins and the *TLC1* RNA), but many groups have successfully produced yeast extracts enriched for telomerase activity (Boule et al., 2005; Cohn & Blackburn, 1995; Nickens et al., 2018, 2019). Though largely undefined, these extracts are useful to determine the effects of various reagents (e.g., recombinant proteins or small molecules) on telomerase extension of model DNA substrates using the telomerase primer extension assay described in the next section.

1. Over-expression of *S. cerevisiae* Est2/*TLC1* via galactose induction. Grow cultures at 30 °C to an OD₆₀₀ of ~1, and add dry galactose to a final concentration of 2%. Incubate cultures at 30 °C for an additional 16 h. Yeast telomerase has also been over-produced approximately 10-fold by co-galactose induction of GFP-Est1, GFP-Est2, Est3, and *TLC1* in the W303 genetic background (Lin et al., 2015), making this an excellent strain to use for this protocol.
2. Harvest the cells by centrifugation at 3000 × *g* for 10 min at 4 °C.
3. Resuspend the cells in 1 mL of L Buffer per mL of cell paste in a 50-mL conical tube, and flash-freeze the cells as “BBs” by puncturing the tube with an 18-gauge needle and dripping the slurry into one or more 50-mL conical tubes containing liquid nitrogen.
4. Lyse the cells by grinding the frozen BBs with dry ice. This step can be accomplished with a mortar and pestle or more rapidly with an inexpensive electric coffee grinder. Only use enough dry ice to keep the cell BBs and lysate powder frozen during grinding because you need to wait for it to sublime away (e.g., overnight at −80 °C) before proceeding to the next step.
5. Slowly thaw the lysate at 4 °C or in an ice water bath, and clarify it by centrifugation at 10,000 × *g*.
6. Pre-equilibrate DEAE-Sephacrose Fast Flow resin (GE Healthcare) with TMG500 Buffer, and incubate it with the clarified cell lysate for 30 min at 4 °C.
7. Gently centrifuge the resin for 1 min at 800 × *g*, decant the buffer, and wash it three times with TMG500 Buffer.
8. Elute telomerase with TMG900 buffer, and concentrate the eluate to ~4 mL using an Amicon Ultra-4 centrifugal filter with a 30,000 Da cutoff.
9. Desalt the telomerase preparation using a 4-mL Zeba spin column (ThermoFisher Scientific), and further concentrate the sample to ~100 µL/L of starting culture.
10. Dilute the samples 1:1 with 100% glycerol, flash-freeze with liquid nitrogen, and store at −80 °C.
11. Each telomerase preparation should be titrated to standardize activity levels before use in experiments (Fig. 3). Control experiments should also be performed to ensure that extension activity is RNase A sensitive.

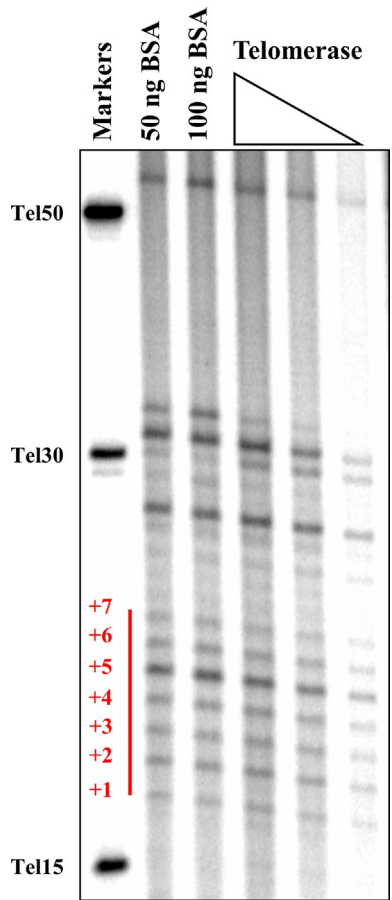


Fig. 3 Titrating a fresh telomerase preparation. The ability of increasing amounts of a telomerase extract to perform primer extension of the Tel15 substrate is tested for each batch of extract prepared to standardize activity levels with previous experiments. The +1 to +7 nucleotide additions are noted in red, and we often see a prominent +5 extension. The markers are ^{32}P -labeled Tel15, Tel30, and Tel50 oligonucleotides. The addition of two amounts of BSA to the primer extension assay was also tested in this experiment to control for the effects of added protein to the reaction.

4.4 Telomerase primer extension assay

This assay utilizes one or more single-stranded DNA substrates comprised of *S. cerevisiae* telomeric repeat sequence (TG_{1-3}) and radiolabeled TTP to measure the reverse transcriptase activity of the yeast telomerase extract produced above.

1. Perform telomerase reactions in 10 μL of $1 \times$ Telomerase Reaction Buffer, and incubate at 30°C for 45 min. In reactions with two primers, each is used at $0.5 \mu\text{M}$. For combined helicase and telomerase assays, the desired helicase is diluted into $1 \times$ Hrql Storage Buffer so that a 10-fold dilution into reactions achieves the desired helicase concentration. Control reactions lacking added helicase receive 1 μL of $1 \times$ Hrql Storage Buffer without added protein. We have found that a range of 0–360 nM Hrql and/or Pif1 is informative.
2. Stop reactions by adding 25 volumes of Stop Buffer and incubating at 30°C for 45 min.
3. Extract samples twice with phenol/chloroform (pH 7.0), and precipitate DNA by the addition of 1 volume of 4 M NH_4OAc and 2.5 volumes of ice-cold 100% ethanol.
4. Chill tubes at -80°C for ≥ 30 min, and centrifuged them at $21,130 \times g$ to pellet reaction products.
5. Resuspend pellets in 6 μL of 95% (v/v) formamide load buffer, heat to 95°C for 5 min, and separate reaction products on 16% 19:1 acrylamide/bisacrylamide gels containing 6 M urea run at 2500 V for 120 min.
6. Dry the gels using a vacuum gel drier, expose the dried gel to a phosphorimager plate for ~ 24 h, and image using a Typhoon 9500 scanner or similar phosphorimager. The gels can also be frozen at -20°C for phosphorimager or film exposure. In either case, to avoid gel cracking during drying or freezing, the gels should be poured with a minimal amount of TEMED (e.g., 10 $\mu\text{L}/100 \text{ mL}$ of degassed acrylamide-TBE solution) and allowed to slowly polymerize.
7. Quantify total activity by measuring the densitometry for each band on a gel using ImageQuant, ImageJ, or similar software. The sum of the measured values for each band in a lane is reported as the total activity. Bands should be corrected for the number of dT residues (i.e., the amount of $[\alpha\text{-}^{32}\text{P}]\text{TTP}$ incorporation) and normalized to the loading control to generate corrected pixel values. The distribution of label is determined by adding the corrected pixels for each individual band to obtain total cumulative activity. Each individual band is then divided by the total activity to determine the percent of label in each band, referred to as label distribution. Telomerase extensions T51–T58 from experiments with the Tel50 substrate are quantified as a group due to low resolution of the bands. Processivity is calculated by submitting the corrected pixel data to the following calculation: $(T_1 + 1 \dots T_n)/(T_1 + \dots T_n)$, where T_1 is the corrected pixel value of the lowest molecular weight telomerase extension band, and T_n is the highest molecular weight band in a series.

References

- Andis, N. M., Sausen, C. W., Alladin, A., & Bochman, M. L. (2018). The WYL domain of the PIF1 helicase from the thermophilic bacterium *Thermotoga elfii* is an accessory single-stranded DNA binding module. *Biochemistry*, 57(7), 1108–1118. <https://doi.org/10.1021/acs.biochem.7b01233>.
- Azam, M., Lee, J. Y., Abraham, V., Chanoux, R., Schoenly, K. A., & Johnson, F. B. (2006). Evidence that the *S. cerevisiae* Sgs1 protein facilitates recombinational repair of telomeres during senescence. *Nucleic Acids Research*, 34, 506–516. 34/2/506 [pii] <https://doi.org/10.1093/nar/gkj452>.
- Bochman, M. L. (2014). Roles of DNA helicases in the maintenance of genome integrity. *Molecular & Cellular Oncology*, 1(3). <https://doi.org/10.4161/23723548.2014.963429>, e963429.
- Bochman, M. L., Paeschke, K., Chan, A., & Zakian, V. A. (2014). Hrq1, a homolog of the human RecQ4 helicase, acts catalytically and structurally to promote genome integrity. *Cell Reports*, 6(2), 346–356. <https://doi.org/10.1016/j.celrep.2013.12.037>.
- Bochman, M. L., Paeschke, K., & Zakian, V. A. (2012). DNA secondary structures: Stability and function of G-quadruplex structures. *Nature Reviews. Genetics*, 13(11), 770–780. <https://doi.org/10.1038/nrg3296>.
- Bochman, M. L., Sabouri, N., & Zakian, V. A. (2010). Unwinding the functions of the Pif1 family helicases. *DNA Repair (Amst)*, 9, 237–249. doi: S1568-7864(10)00011-X [pii] <https://doi.org/10.1016/j.dnarep.2010.01.008>.
- Bonnell, E., Pasquier, E., & Wellinger, R. J. (2021). Telomere replication: Solving multiple end replication problems. *Frontiers in Cell and Development Biology*, 9, 668171. <https://doi.org/10.3389/fcell.2021.668171>.
- Boule, J., Vega, L., & Zakian, V. (2005). The Yeast Pif1p helicase removes telomerase from DNA. *Nature*, 438, 57–61. Retrieved from <http://www.nature.com/nature/journal/v438/n7064/abs/nature04091.html>.
- Casjens, S. (1999). Evolution of the linear DNA replicons of the *Borrelia* spirochetes. *Current Opinion in Microbiology*, 2(5), 529–534. [https://doi.org/10.1016/s1369-5274\(99\)00012-0](https://doi.org/10.1016/s1369-5274(99)00012-0).
- Cohen, H., & Sinclair, D. (2001). Recombination-mediated lengthening of terminal telomeric repeats requires the Sgs1 DNA helicase. *Proceedings of the National Academy of Sciences of the United States of America*, 98, 3174–3179.
- Cohn, M., & Blackburn, E. H. (1995). Telomerase in yeast. *Science*, 269, 396–400.
- Huang, P., Pryde, F., Lester, D., Maddison, R., Borts, R., Hickson, I., et al. (2001). SGS1 is required for telomere elongation in the absence of telomerase. *Current Biology*, 11, 125–129.
- Ivessa, A. S., Zhou, J.-Q., Schulz, V. P., Monson, E. M., & Zakian, V. A. (2002). *Saccharomyces* Rrm3p, a 5' to 3' DNA helicase that promotes replication fork progression through telomeric and sub-telomeric DNA. *Genes & Development*, 16, 1383–1396.
- Johnson, F., Marciniak, R., McVey, M., Stewart, S., Hahn, W., & Guarente, L. (2001). The *Saccharomyces cerevisiae* WRN homolog Sgs1p participates in telomere maintenance in cells lacking telomerase. *The EMBO Journal*, 20, 905–913.
- Kobryn, K., & Chaconas, G. (2001). The circle is broken: Telomere resolution in linear replicons. *Current Opinion in Microbiology*, 4(5), 558–564. [https://doi.org/10.1016/s1369-5274\(00\)00251-4](https://doi.org/10.1016/s1369-5274(00)00251-4).
- Lin, K. W., McDonald, K. R., Guise, A. J., Chan, A., Cristea, I. M., & Zakian, V. A. (2015). Proteomics of yeast telomerase identified Cdc48–Npl4–Ufd1 and Ufd4 as regulators of Est1 and telomere length. *Nature Communications*, 6, 8290. <https://doi.org/10.1038/ncomms9290>.
- Nickens, D. G., Rogers, C. M., & Bochman, M. L. (2018). The *Saccharomyces cerevisiae* Hrq1 and Pif1 DNA helicases synergistically modulate telomerase activity in vitro. *The Journal of Biological Chemistry*, 293(37), 14481–14496. <https://doi.org/10.1074/jbc.RA118.004092>.

- Nickens, D. G., Sausen, C. W., & Bochman, M. L. (2019). The biochemical activities of the *Saccharomyces cerevisiae* Pif1 helicase are regulated by its N-terminal domain. *Genes (Basel)*, 10(6). <https://doi.org/10.3390/genes10060411>.
- Ononye, O. E., Sausen, C. W., Balakrishnan, L., & Bochman, M. L. (2020). Lysine acetylation regulates the activity of nuclear Pif1. *The Journal of Biological Chemistry*. <https://doi.org/10.1074/jbc.RA120.015164>.
- Ramanagoudr-Bhojappa, R., Blair, L. P., Tackett, A. J., & Raney, K. D. (2013). Physical and functional interaction between yeast Pif1 helicase and Rim1 single-stranded DNA binding protein. *Nucleic Acids Research*, 41(2), 1029–1046. <https://doi.org/10.1093/nar/gks1088>.
- Rogers, C. M., & Bochman, M. L. (2017). *Saccharomyces cerevisiae* Hrq1 helicase activity is affected by the sequence but not the length of single-stranded DNA. *Biochemical and Biophysical Research Communications*, 486(4), 1116–1121. <https://doi.org/10.1016/j.bbrc.2017.04.003>.
- Rogers, C. M., Wang, J. C., Noguchi, H., Imasaki, T., Takagi, Y., & Bochman, M. L. (2017). Yeast Hrq1 shares structural and functional homology with the disease-linked human RecQ4 helicase. *Nucleic Acids Research*, 45(9), 5217–5230. <https://doi.org/10.1093/nar/gkx151>.
- Smith, J. S., Chen, Q., Yatsunyk, L. A., Nicoludis, J. M., Garcia, M. S., Kranaster, R., et al. (2011). Rudimentary G-quadruplex-based telomere capping in *Saccharomyces cerevisiae*. *Nature Structural & Molecular Biology*, 18(4), 478–485. <https://doi.org/10.1038/nsmb.2033>.
- Studier, F. W. (2005). Protein production by auto-induction in high density shaking cultures. *Protein Expression and Purification*, 41(1), 207–234. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/15915565>.
- Volff, J. N., & Altenbuchner, J. (2000). A new beginning with new ends: Linearisation of circular chromosomes during bacterial evolution. *FEMS Microbiology Letters*, 186(2), 143–150. <https://doi.org/10.1111/j.1574-6968.2000.tb09095.x>.
- Wellinger, R. J., & Zakian, V. A. (2012). Everything you ever wanted to know about *Saccharomyces cerevisiae* telomeres: Beginning to end. *Genetics*, 191(4), 1073–1105. <https://doi.org/10.1534/genetics.111.137851>.
- Yang, C. C., Huang, C. H., Li, C. Y., Tsay, Y. G., Lee, S. C., & Chen, C. W. (2002). The terminal proteins of linear *Streptomyces* chromosomes and plasmids: A novel class of replication priming proteins. *Molecular Microbiology*, 43(2), 297–305. Retrieved from <https://www.ncbi.nlm.nih.gov/pubmed/11985710>.
- Zhao, S., Douglas, N. W., Heine, M. J., Williams, G. M., Winther-Larsen, H. C., & Meaden, P. G. (1994). The STL1 gene of *Saccharomyces cerevisiae* is predicted to encode a sugar transporter-like protein. *Gene*, 146(2), 215–219. [https://doi.org/10.1016/0378-1119\(94\)90295-x](https://doi.org/10.1016/0378-1119(94)90295-x).